

Singularitarianism

Singularitarianism is a movement defined by the belief that a technological singularity—the creation of superintelligence—will likely happen in the medium future, and that deliberate action ought to be taken to ensure that the singularity benefits humans.^[1]

Singularitarians are distinguished from other futurists who speculate on a technological singularity by their belief that the singularity is not only possible, but desirable if guided prudently. Accordingly, they may sometimes dedicate their lives to acting in ways they believe will contribute to its rapid yet safe realization.^[2]

American news magazine *Time* describes the worldview of Singularitarians by saying "even though it sounds like science fiction, it isn't, no more than a weather forecast is science fiction. It's not a fringe idea; it's a serious hypothesis about the future of life on Earth. There's an intellectual gag reflex that kicks in anytime you try to swallow an idea that involves super-intelligent immortal cyborgs, but... while the Singularity appears to be, on the face of it, preposterous, it's an idea that rewards sober, careful evaluation".^[1]

Definition

The term "Singularitarian" was originally defined by Extropian thinker Mark Plus (Mark Potts) in 1991 to mean "one who believes the concept of a Singularity".^[3] This term has since been redefined to mean "Singularity activist" or "friend of the Singularity"; that is, one who acts so as to bring about the singularity.^[4]

Singularitarianism can also be thought of as an orientation or an outlook that prefers the enhancement of human intelligence as a specific transhumanist goal instead of focusing on specific technologies such as A.I.^[5] There are also definitions that identify a singularitarian as an activist or a friend of the concept of singularity, that is, one who acts so as to bring about a singularity.^[6] Some sources described it as a moral philosophy that advocates deliberate action to bring about and steer the development of a superintelligence that will lead to a theoretical future point that emerges during a time of accelerated change.^[7]

Inventor and futurist Ray Kurzweil, author of the 2005 book *The Singularity Is Near: When Humans Transcend Biology*, defines a Singularitarian as someone "who understands the Singularity and who has reflected on its implications for his or her own life"^[2] and estimates the singularity will occur around 2045.^[2]

History

An early singularitarian articulation that history is making progress toward a point of superhuman intelligence is found in Hegel's work *The Phenomenology of Spirit*.^[8] In 1993, mathematician, computer scientist, and science fiction author Vernor Vinge hypothesized that the moment might come when technology will allow "creation of entities with greater than human intelligence"^[9] and used the term "the Singularity" to describe this moment.^[10] He suggested that the singularity may pose an existential risk for humanity, and that it could happen through one of four means:

1. The development of computers that are "awake" and superhumanly intelligent.
2. Large computer networks (and their associated users) may "wake up" as a superhumanly intelligent entity.
3. Computer/human interfaces may become so intimate that users may reasonably be considered superhumanly intelligent.
4. Biological science may find ways to improve upon the natural human intellect.^[11]

Singularitarianism coalesced into a coherent ideology in 2000, when artificial intelligence (AI) researcher Eliezer Yudkowsky wrote *The Singularitarian Principles*,^{[2][12]} in which he states that a Singularitarian believes that the singularity is a secular, non-mystical event that is possible, beneficial to the world, and worked toward by its adherents.^[12] Yudkowsky's definition is inclusive of various interpretations.^[5] Theorists such as Michael Anissimov argue for a strict definition that refers only to the advocacy of the development of superintelligence.^[5]

In June 2000, Yudkowsky, with the support of Internet entrepreneurs Brian Atkins and Sabine Atkins, founded the Machine Intelligence Research Institute to work toward the creation of self-improving Friendly AI. MIRI's writings that an AI with the ability to improve upon its own design (Seed AI) would rapidly lead to superintelligence. These Singularitarians believe that reaching the singularity swiftly and safely is the best possible way to minimize net existential risk.

Many people believe a technological singularity is possible without adopting Singularitarianism as a moral philosophy. Although the exact numbers are hard to quantify, Singularitarianism is a small movement, which includes transhumanist philosopher Nick Bostrom. Inventor and futurist Ray Kurzweil, who predicts that the Singularity will occur circa 2045, greatly contributed to popularizing Singularitarianism with his 2005 book *The Singularity Is Near: When Humans Transcend Biology*.^[2]

What, then, is the Singularity? It's a future period during which the pace of technological change will be so rapid, its impact so deep, that human life will be irreversibly transformed. Although neither utopian or dystopian, this epoch will transform the concepts we rely on to give meaning to our lives, from our business models to the cycle of human life, including death itself. Understanding the Singularity will alter our perspective on the significance of our past and the ramifications for our future. To truly understand it inherently changes one's view of life in general and one's particular life. I regard someone who understands the Singularity and who has reflected on its implications for his or her own life as a "singularitarian."^[2]

With the support of NASA, Google, and a broad range of technology forecasters and technocapitalists, the Singularity University opened in 2009 at the NASA Research Park in Silicon Valley with the goal of preparing the next generation of leaders to address the challenges of accelerating change.

In July 2009, many prominent Singularitarians participated in a conference organized by the Association for the Advancement of Artificial Intelligence (AAAI) to discuss the potential impact of robots and computers and the possibility that they may become self-sufficient and able to make their own decisions. They discussed the possibility and the extent to which computers and robots might be able to acquire autonomy, and to what degree they could use such abilities to pose a threat or hazard (i.e., cybernetic revolt). They noted that some machines have acquired various forms of semi-autonomy, including being able to find power sources on their own and independently choose targets to attack with weapons. They warned that some computer viruses can evade elimination and have achieved "cockroach intelligence". They asserted that self-awareness as depicted in science fiction is probably unlikely, but that there are other potential hazards and pitfalls.^[10] Some experts and academics have questioned the use of robots for military combat, especially when such robots are given some degree of autonomous functions.^[13] The President of the AAAI has commissioned a study of this issue.^[14]

Reception

There are several objections to Kurzweil's singularitarianism, even from optimists in the A.I. field. For instance, Pulitzer Prize-winning author Douglas Hofstadter argued that Kurzweil's predicted achievement of human-level A.I. by 2045 is not viable.^[15] Even Gordon Moore, the namesake of Moore's Law that predicated^[16] the notion of singularity, maintained that it will never occur.^[17] According to some observers, these criticisms do not diminish enthusiasm for singularity because it has assumed a quasi-religious response to the fear of death, allowing its adherents to enjoy the benefits of religion without its ontological burdens.^[15] Science journalist John Horgan wrote:

Let's face it. The singularity is a religious rather than a scientific vision. The science-fiction writer Ken MacLeod has dubbed it "the rapture for nerds," an allusion to the end-time, when Jesus whisks the faithful to heaven and leaves us sinners behind. Such yearning for transcendence, whether spiritual or technological, is all too understandable. Both as individuals and as a species, we face deadly serious problems, including terrorism, nuclear proliferation, overpopulation, poverty, famine, environmental degradation, climate change, resource depletion, and AIDS. Engineers and scientists should be helping us face the world's problems and find solutions to them, rather than indulging in escapist, pseudoscientific fantasies like the singularity.^[18]

Kurzweil rejects this assessment, saying that his predictions about the singularity are driven by the data that increases in computational technology have long been exponential.^[19] He says that his critics mistakenly take an intuitive, linear view of technological advancement rather than accounting for that exponential growth.^[20]

See also

- [AI mysticism](#)
- [Artificial general intelligence](#)
- [Eschatology](#)
- [Existential risk from artificial general intelligence](#)
- [Global brain](#)
- [Intelligence explosion](#)
- [Outline of transhumanism](#)
- [Post-scarcity economy](#)
- [Technological utopianism](#)

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External links

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- "The Consciousness Conundrum" (<https://spectrum.ieee.org/the-consciousness-conundrum>), a criticism of singularitarians by John Horgan

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